

PHYS 1408 CHAPTER 13 BIG G GRAVITY

Principia 1687 Isaac Newton

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$$F = \frac{G M_1 M_2}{r^2} \quad G = 6.67 \cdot 10^{-11} \frac{\text{Nm}^2}{\text{kg}^2}$$

$$M_{\text{Earth}} = 5.97 \cdot 10^{24} \text{ kg}$$

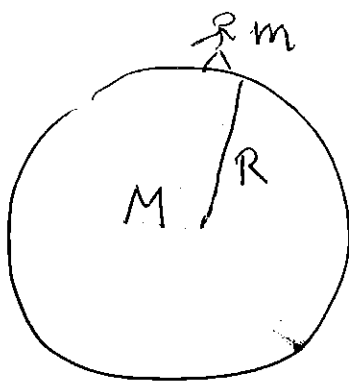
$$R_{\text{Earth}} = 6.37 \cdot 10^6 \text{ m}$$

$$M_{\text{Sun}} = 2 \cdot 10^{30} \text{ kg}$$

$$M_{\text{Moon}} = M_{\text{Earth}} / 80$$

$$d_{ES} = 1.5 \cdot 10^{11} \text{ m} \quad 150,000,000 \text{ km} \quad 93,000,000 \text{ miles}$$

$$d_{EM} = 4 \cdot 10^8 \text{ m} \quad 400,000 \text{ km} \quad 240,000 \text{ miles}$$



$$F = mg = \frac{G M m}{R^2}$$

$$g = \frac{G M_E}{R_E^2} = \frac{6.67 \cdot 10^{-11} \cdot 5.97 \cdot 10^{24}}{(6.37 \cdot 10^6)^2}$$

$$= 9.81 \frac{\text{m}}{\text{s}^2}$$

Satellite orbiting a planet

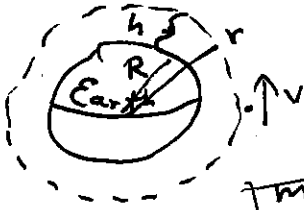
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$$r = R + h$$

r radius of satellite's orbit about center of earth

h altitude above earth's surface

R radius of earth



m satellite mass

FBD



$$\downarrow \frac{GMm}{r^2}$$

$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$\frac{GM}{r} = v^2$$

$$v = \frac{2\pi r}{T}$$

$2\pi r$ distance around planet

T period

$$\frac{GM}{r} = \left(\frac{2\pi r}{T}\right)^2 = \frac{4\pi^2 r^3}{T^2}$$

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$\text{if } h = 400 \text{ km} = 4 \cdot 10^5 \text{ m}$$

$$r = R + h = 6.37 \cdot 10^6 + 4 \cdot 10^5 \text{ m}$$

$$= 6.77 \cdot 10^6 \text{ m}$$

$$T = \sqrt{\frac{4\pi^2 (6.77 \cdot 10^6)^3}{6.67 \cdot 10^{-11} \cdot 5.97 \cdot 10^{24}}} = 5546 \text{ s} = 92.4 \text{ minutes}$$

LOW Earth Orbit

LEO

Geosynchronous orbit

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(Earth with time)

24hr orbit places

equatorial satellite over same
point on the equator.

$$T = 24\text{hr} = 8.64 \cdot 10^4 \text{ s}$$

$$T^2 = \frac{4\pi^2 r^3}{GM}$$

$$r^3 = \frac{GM T^2}{4\pi^2}$$

$$r = \left(\frac{T^2 GM}{4\pi^2} \right)^{1/3}$$

$$r = \left(\frac{(8.64 \cdot 10^4)^2 \cdot 6.67 \cdot 10^{-11} \cdot 5.97 \cdot 10^{24}}{4\pi^2} \right)^{1/3}$$

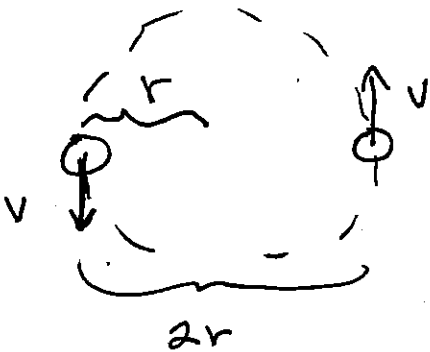
$$r = 4.22 \cdot 10^7 \text{ m}$$

$$h = r - R = 4.22 \cdot 10^7 - 6.37 \cdot 10^6 = 3.59 \cdot 10^7 \text{ m}$$

$$= 35,900 \text{ km} \quad (\sim 22,400 \text{ miles})$$

2 equal mass stars

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$$M_1 = M_2 = M_{\text{sun}} = 2 \cdot 10^{30} \text{ kg}$$

$$r = 6 \cdot 10^{10} \text{ m} \sim \text{Sun-Mercury distance}$$

$$\frac{GM^2}{(2r)^2} = \frac{Mv^2}{r}$$

$$v = \frac{2\pi r}{T}$$

$$\frac{GM}{4r} = \frac{4\pi^2 r^3}{T^2}$$

$$T^2 = \frac{16\pi^2 r^3}{GM}$$

$$T = \sqrt{\frac{16\pi^2 (6 \cdot 10^{10})^3}{6.67 \cdot 10^{-11} \cdot 2 \cdot 10^{30}}}$$

$$= 1.599 \cdot 10^7 \text{ s}$$

$$= 2.67 \cdot 10^5 \text{ min}$$

$$= 4441 \text{ hr}$$

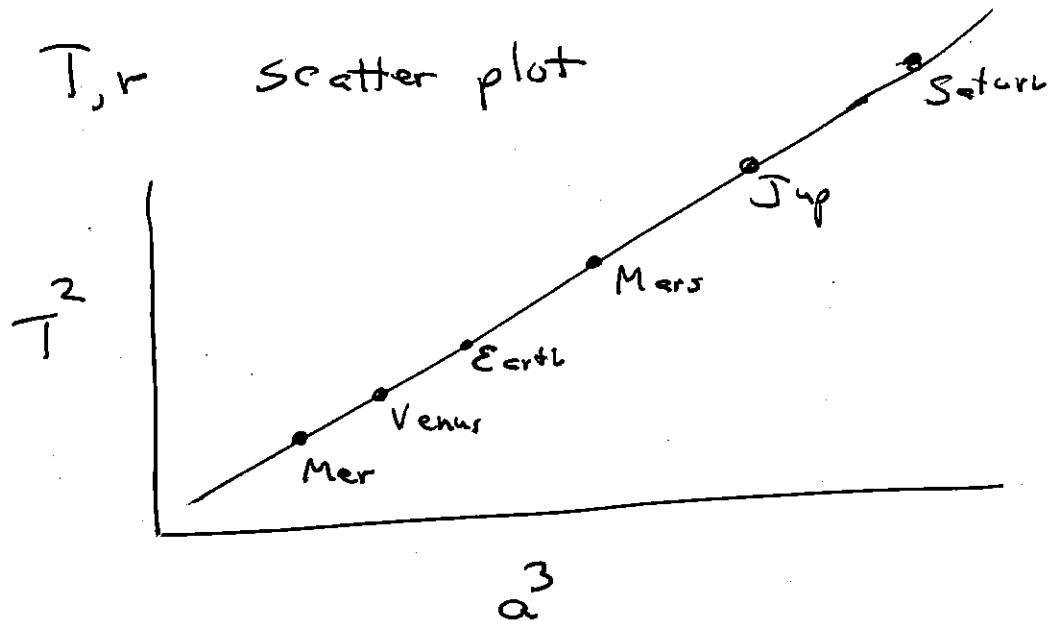
$$= 185 \text{ days}$$

separation btwn the masses is $2r$ (not r)

Johannes Kepler (1605)

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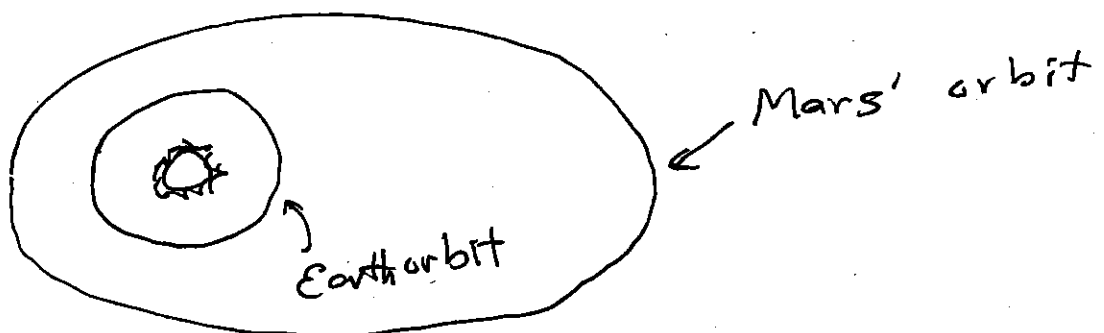
T, r scatter plot



Kepler studied orbit of Mars 20 yrs
Catholic Church Geocentric Universe
Protestants Heliocentric Universe

Both constant velocity circular paths

Kepler found Mars follows elliptical orbit



Copernican Revolution

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Niccolò Copernicus 1473 - 1543

Tycho Brahe 1546 - 1601

Johannes Kepler 1571 - 1630

Galileo Galilei 1564 - 1642

Isaac Newton 1642 - 1727

* Before Copernicus man thought
the cosmos swung around
Earth.

* After Newton observational science
led straight to us today.

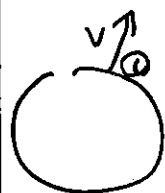
Gravitational Potential Energy

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$$U = - \frac{GMm}{r}$$

Rocket launch $v_i = 1000 \text{ m/s}$ $h = ?$

CGE ③ $v=0$ $U_1 + K_1 = U_2$ $K_2 = 0$ $U_1 = -\frac{GMm}{R}$
 $h = ?$
 $r = ?$



$$-\frac{GMm}{R} + \frac{1}{2}mv_i^2 = -\frac{GMm}{r}$$

$$r = R + h$$

$$\frac{v_i^2}{2} - \frac{GM}{R} = -\frac{GM}{r}$$

$$\frac{GM}{r} = \frac{GM}{R} - \frac{v_i^2}{2}$$

~~$$\frac{GM}{R} = \frac{GM}{r} + \frac{v_i^2}{2}$$~~

$$\frac{1}{r} = \frac{1}{R} - \frac{v_i^2}{2GM}$$

~~$$\frac{1}{R} = \frac{1}{r} + \frac{v_i^2}{2GM}$$~~

$$r = \frac{1}{\frac{1}{R} - \frac{v_i^2}{2GM}} = R + h$$

$$h = \frac{1}{\frac{1}{R} - \frac{v_i^2}{2GM}} - R$$

$$= \frac{2GMR}{2GM - v_i^2 R} - R$$

$$h = \frac{2 \cdot 6.67 \cdot 10^{-11} \cdot 5.97 \cdot 10^{24} \cdot 6.37 \cdot 10^6}{2 \cdot 6.67 \cdot 10^{-11} \cdot 5.97 \cdot 10^{24} - (10^3)^2 \cdot 6.37 \cdot 10^6} - 6.37 \cdot 10^6 \quad \frac{8}{10}$$

$$h = \frac{5.07 \cdot 10^{21}}{7.96 \cdot 10^{14} - 6.37 \cdot 10^{12}} - 6.37 \cdot 10^6 = 5.073 \cdot 10^4 \text{ m}$$

$$h = 50.73 \text{ km}$$

What if given a target altitude? $h = 1000 \text{ km}$

what v to launch?

$$\frac{v^2}{2} - \frac{GM}{R} = -\frac{GM}{r}$$

$$h = 10^6 \text{ m}$$

$$r = h + R = 7.37 \cdot 10^6 \text{ m}$$

$$\frac{v^2}{2} = GM \left(\frac{1}{R} - \frac{1}{r} \right)$$

$$v = \sqrt{2GM \left(\frac{1}{R} - \frac{1}{r} \right)} = \sqrt{2 \cdot 6.67 \cdot 10^{-11} \cdot 5.97 \cdot 10^{24} \left(\frac{1}{6.37 \cdot 10^6} - \frac{1}{7.37 \cdot 10^6} \right)}$$

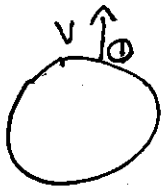
$$v = 4119 \text{ m/s}$$

V_{escape}

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② $r = \infty$
 $v = 0$ $E_1 = E_2$

$$U_1 + K_1 = U_2 + K_2$$



$$-\frac{GMm}{R} + \frac{mv^2}{2} = 0 + 0$$

$$\frac{GM}{R} = \frac{v^2}{2}$$

$$v = \sqrt{\frac{2GM}{R}} = \sqrt{\frac{2 \cdot 6.67 \cdot 10^{-11} \cdot 5.97 \cdot 10^{24}}{6.37 \cdot 10^6}}$$

$$= 11,181 \text{ m/s} \sim 11.2 \text{ km/s}$$

if $v_{\text{escape}} = c$ speed of light $3 \cdot 10^8 \text{ m/s}$

then r is the radius of the event horizon

$$c^2 = \frac{2GM}{r}$$

$$r = \frac{2GM}{c^2}$$

if mass of the
~~galaxy is $2 \cdot 10^{11} M_{\odot}$~~

$$\approx 4 \cdot 10^{41} \text{ kg}$$

$$5 \cdot 10^6 M_{\odot} = 10 \cdot 10^{36}$$

$$14 = 10^{37} \text{ kg}$$

50 million solar
masses

$$r = \frac{2 \cdot 6.67 \cdot 10^{-11} \cdot 10^{37}}{(3 \cdot 10^8)^2}$$

$$= 1.48 \cdot 10^{10} \text{ m}$$

$\frac{1}{10}$ radius of Earth's orbit

Orbital Speed of planets

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$$\frac{GMm}{r^2} = \frac{mv^2}{r}$$

$$1 \text{ AU} = 1.5 \cdot 10^{11} \text{ m}$$

$$M_{\text{Sun}} = 2 \cdot 10^{30} \text{ kg}$$

$$v = \sqrt{\frac{GM}{r}}$$

r Planet	R	R (m)	$v = \sqrt{\frac{1.33 \cdot 10^{20}}{r}}$
Merc	0.4	$0.6 \cdot 10^{11}$	47.15 km/s
Venus	0.7	$1.05 \cdot 10^{11}$	35.59 km/s
Earth	1 AU	$1.5 \cdot 10^{11}$	29.78 km/s
Mars	1.5	$2.25 \cdot 10^{11}$	24.31 km/s
Jupiter	5.2	$7.8 \cdot 10^{11}$	13.06 km/s
Saturn	10	$1.5 \cdot 10^{12} \text{ m}$	9.42 km/s
Uranus	20	$3 \cdot 10^{12} \text{ m}$	6.66 km/s
Neptune	30	$4.5 \cdot 10^{12} \text{ m}$	5.44 km/s

Only a factor of 10 in speed of planet!

Way smaller than I expected!